Practical 8: Customer Purchase Prediction Using Decision Tree and Regression Tree

Question
A retail company wants to use customer and sales information for predictive analysis.

Part A – Decision Tree Classification
The company wants to predict whether a customer will purchase a product based on:
Age
Annual Income
Website Visits
Previous Purchases
where:
Purchase = 1 → Yes
Purchase = 0 → No

Using the dataset below, build a Decision Tree Classifier and predict the purchase decision.

Dataset:

Customer	Age	Annual income(in lakhs)	Website Visits	Previous Purchases	Purchase
C1	22	25	2	1	0
C2	25	30	3	1	0
C3	28	35	4	2	0
C4	30	40	5	2	1
C5	32	45	6	3	1
C6	35	50	7	3	1
C7	38	55	8	4	1
C8	40	60	9	4	1
C9	42	65	10	5	1
C10	24	28	2	1	0
C11	27	32	3	1	0
C12	31	42	5	2	1

C13	36	52	7	3	1
C14	29	70	9	5	1
C15	41	38	4	2	0

Part B – Regression Tree
A company wants to predict Sales based on:
Advertising expenditure
Number of sales employees
Store size

Build a Regression Tree (CART) and predict sales for a new store with:
Advertising = ₹7.5 lakhs
Sales Employees = 11
Store Size = 2400 sq. ft.
Evaluate the model using R^2 score.

Solutions:

Part A – Decision Tree Classifier

Step 1: Import Libraries

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
```

Step 2: Create Dataset

```
data = {
   "Age": [22,25,28,30,32,35,38,40,42,24,27,31,36,45,29],
   "Income": [25,30,35,40,45,50,55,60,65,28,32,42,52,70,38],
   "Website_Visits": [2,3,4,5,6,7,8,9,10,2,3,5,7,9,4],
   "Previous_Purchases": [1,1,2,2,3,3,4,4,5,1,1,2,3,5,2],
   "Purchase": [0,0,0,1,1,1,1,1,1,0,0,1,1,1,0]
}

df = pd.DataFrame(data)

print(df)
```

Step 3: Define X and y

```
X = df[[
    "Age",
    "Income",
    "Website_Visits",
    "Previous_Purchases"
]]

y = df["Purchase"]

Step 4: Split the Data
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

Step 5: Create Decision Tree
model = DecisionTreeClassifier(
    criterion="gini",
    max_depth=3,
    random_state=42
)

Step 6: Train the Model
model.fit(X_train, y_train)

Step 7: Predict Test Data
y_pred = model.predict(X_test)

Step 8: Calculate Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

Step 9: Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)

Step 10: Predict a New Customer

Suppose a new customer has:
Age = 34
Income = ₹48 lakhs
```

```
Website Visits = 6
Previous Purchases = 3

new_customer = pd.DataFrame({
    "Age": [34],
    "Income": [48],
    "Website_Visits": [6],
    "Previous_Purchases": [3]
})

prediction = model.predict(new_customer)

print("Predicted Purchase:", prediction[0])

PART B – Regression Tree

Step 1: Create Dataset
data = {
    "Advertising": [2,3,4,5,6,7,8,9,10,11],
    "Sales_Employees": [5,6,7,8,9,10,11,12,13,14],
    "Store_Size": [1000,1200,1500,1800,2000,
                   2200,2500,2800,3000,3200],
    "Sales": [25,30,38,45,52,60,68,75,84,92]
}

df = pd.DataFrame(data)

Step 2: Define X and y
X = df[[
    "Advertising",
    "Sales_Employees",
    "Store_Size"
]]

y = df["Sales"]

Step 3: Split Data
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

Step 4: Create Regression Tree
```

```
from sklearn.tree import DecisionTreeRegressor

model = DecisionTreeRegressor(
    max_depth=3,
    random_state=42
)

Step 5: Train
model.fit(X_train, y_train)

Step 6: Predict Test Data
y_pred = model.predict(X_test)
print("Predicted Sales:")
print(y_pred)

Step 7: Calculate R²
from sklearn.metrics import r2_score
r2 = r2_score(y_test, y_pred)
print("R² Score:", r2)

Predict Sales for a New Store
The new store has:
Advertising = 7.5
Sales Employees = 11
Store Size = 2400

new_store = pd.DataFrame({
    "Advertising": [7.5],
    "Sales_Employees": [11],
    "Store_Size": [2400]
})

prediction = model.predict(new_store)

print("Predicted Sales:", prediction[0])
```

Output:

Result:

Part A

The Decision Tree Classifier was successfully developed to predict whether customers would purchase a product based on customer characteristics.

Part B

The Regression Tree (CART) was successfully developed to predict sales based on advertising expenditure, number of sales employees, and store size. The model was evaluated using R^2.